PATTERN_QNU_RR: 140-label pooled 140·Q(v) mod 8 vs MF2

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RESULTS: PATTERN_QNU_RR (Map-PQR1)

PREREG sha256 1858b03e721ebfe5ecfb2c43f127a89409b66585744b2fd7bc9264b44965b7d5. Hashed after explore and before test. test.py recomputes the result in exact Fractions and matches explore.

fibre	ρ_res (MF2)	ρ_mod (140-label pool, 140·Q mod 8)
0	1/32	1/8
1	1/4	0
2	3/32	1/4
3	0	1/4
4	7/32	1/8
5	1/4	0
6	5/32	0
7	0	1/4

- PF1 escaped: 5 fibres carry mass.
- P1 (exact equality): false. TV = 3/4.
- ρ_mod puts mass 1/2 on {3,7}, where ρ_res is 0, and misses 1, 5 and 6.
- Non-vacuous.

Verdict: FAIL (Category A). Bridge: **NO.** ## NOTES: PATTERN_QNU_RR ## Explore - N = 140 is the least integral scale and matches the written e(Nz/140) of rev8 l.896. - 140Q mod 8 has period 8 in n (checked against 16). - Per-label supports: 35 labels give {0,2,4} and 35 give {3,7}. - Pools: - all 140: (1/8,0,1/4,1/4,1/8,0,0,1/4) (COMMITTED); - the 28 labels with 70α1 ∈ {7,21,49,63}: (0,0,0,1/2,0,0,0,1/2).

Lemma PQ1 (Category A; exact enumeration over a full period, logs/test.log)

On the coset $\alpha \in L'$ (the 35 α with no a-shift), $140 \cdot Q(\alpha+n) \bmod 8 \in \{0,2,4\}$. On $\alpha \in a+L'$, $140 \cdot Q(\alpha+n) \equiv 3 \bmod 4$, i.e. it lies in {3,7} ($140 \cdot Q(a) = -49 \equiv 7$). Hence for **any** non-negative weighting of the 140 labels (or of β), with scale 140, fibres 1, 5 and 6 get zero mass. ρ_res has 5/8 of its mass there, so no label pool of the binary exponent can equal MF2.

Observation (post hoc, NOT a claim, not preregistered)

The a-coset fills exactly {3,7}, the two fibres that MF2 leaves empty, and the L′ coset sits in {0,2,4}. The two {0,2,4,…}/{3,7} splits look complementary. Testing any transform built from this (e.g. a complement or shift of fibres) would be chosen after seeing ρ_res, so it is fitting. It can only be preregistered if a written reason for that transform exists.

Next pattern step (not run)

Unsigned pools are exhausted (PF1, PQ1). The remaining written structure in the same lift is the phase $e(B(\nu,\beta))$, $\beta \in \{0,-a\}$, together with the signed cone weight $w(\nu)$. This is the signed mass per fibre: $\Sigma\, w(\nu) \cdot e(B(\nu,\beta))$ binned by 140Q mod 8. That is the actual coefficient content of $u_{\alpha,\beta}$, not lattice-point counting. A next package would preregister |signed fibre mass|, normalized, over the 140 labels vs ρ_res. Its density convention (cone-weighted boxes, written w) must be fixed in PREREG.